

An open flame and smoke detection dataset for deep learning in remote sensing based fire detection

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ABSTRACT

The early warning of fires is pivotal for preventing substantial economic losses and ecological damage. However, the enhancement of fire model generalization performance still faces challenges due to limitations in existing fire datasets, such as image quantity and heterogeneity. Taking the advantage of rapid advancements in remote sensing technology and artificial intelligence, we introduce the Flame And Smoke Detection Dataset (FASDD), a pioneering collection comprising over 120,000 heterogeneous images covering various fire scenarios, aiming to promote the evolution of fire detection models. FASDD serves as a challenging benchmark for object detection tasks and features three sub-datasets: FASDD_CV, FASDD_UAV, and FASDD_RS. These sub-datasets comprise cross-domain images from ground-based, airborne, and spaceborne sensors, respectively. Extensive experiments using advanced Swin Transformer models demonstrate satisfactory fire detection performance, with mAP scores of 84.9%, 89.7%, and 74.0% on the respective sub-datasets. Notably, the application case in wildfire localization highlights the proficiency of FASDD_RS models for monitoring and tracking forest fires. Trained on FASDD_RS, FASDD_UAV, and FASDD_CV, deep learning models can be individually deployed on edge devices including satellites, drones, and terrestrial sensors, thus empowering collaborative fire warning within the space-air-ground integrated observation network. Additionally, a standardized procedure for data preparation, data procedure, and quality assurance is established, providing off-the-shelf annotations with four common configurations to facilitate machine learning model training. FASDD stands as a pivotal benchmark, providing valuable resources for training and evaluating deep learning fire detection models, especially Large Vision Models (LVM). It holds the potential to accelerate research progress in urban firefighting or forest fire detection, providing vital security measures for early fire warning and emergency response tasks.

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1. Introduction

Fire constitutes a significant hazard to both human well-being and ecological balance (Gibson et al. 2020), resulting in considerable economic losses and ecological harm while jeopardizing lives (Chowdary, Gupta, and Singh 2018; Gaur et al. 2020). Existing research indicates a growing trend in the frequency and duration of natural disasters such as extreme weather events or wildfires on a global scale (Jones et al. 2022; Tzavella, Skopeliti, and Fekete 2024). The 2019 Australian bushfire crisis serves as a prime example of extreme fire disasters, underscoring the critical need for fire detection in the pre-suppression phase. Various systems, including ground-based, airborne, or spaceborne platforms, outfitted with visible light, thermal infrared, and multispectral instruments, have been designed and widely applied. Single-sensor fire

detection often grapples with issues like sensitivity or false alarms, prompting suggested remedies such as data fusion and the establishment of collaborative observation networks (Li et al. 2023). Strategies involving data fusion and collaborative observation networks have been proposed specifically. Numerous camera networks, integrating Computer Vision (CV), and machine learning, have been put into the operation, contributing to the improvement of fire detection and localization (Barmpoutis et al. 2020; Govil et al. 2020). Nevertheless, under intricate real-world scenarios, flame or smoke displays various attributes, including flickering, gradual growth, irregularity, color variation, and various light intensities (Muhammad, Ahmad, and Baik 2018). Flame can be easily misinterpreted as lights, the sun, or leaves, while smoke may bear a resemblance to clouds, hair, or waterfalls

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(Chino et al. 2015; Ko, Kwak, and Nam 2012). These intricacies, compounded by the real-world scenario's low signal-to-noise ratio, introduce further challenges to vision-based algorithms designed to detect fires.

Vision-based algorithms for detecting fires can be broadly categorized into static/dynamic feature-driven, machine learning-driven, as well as deep learning-driven approaches. Typically, static feature-driven techniques carry out the discrimination of fires by leveraging key characteristics of flame or smoke, like color or texture (Calderara, Piccinini, and Cucchiara 2008). Despite their lower computational costs, these methods exhibit reduced reliability and an increased incidence of false alarms. Dynamic feature-driven methods, on the other hand, analyze flame and smoke videos by considering factors such as flicker, motion, and dynamic texture (Dimitropoulos, Barmpoutis, and Grammalidis 2015; Töreyn, Dedeoğlu, and Çetin 2005). These methods leverage the inherent changing characteristics of flame and smoke, such as growth and disorder, to improve fire detection performance to a certain extent, thereby providing support for disaster assessment and emergency response (Zhu et al. 2024). Yet, they come with the trade-off of requiring higher computational costs. Traditional fire detection methods based on machine learning, such as those employing random forest, support vector machine, and decision tree algorithms, rely on the utilization of manually crafted features during training (Chi, Lu, and Ji 2017; S. Wang et al. 2017). However, these algorithms encounter challenges associated with feature selection bias, and operational complexity, as well as time costs, are generally high.

As Artificial Intelligence (AI) rapidly advances in the Remote Sensing (RS) domain (Cao et al. 2023; Z. Zhang et al. 2023), the deep learning-driven methodologies have progressively gained prominence in fire detection research (Nadeem et al. 2023; J. Yuan et al. 2023). Dua et al. (2020) proposed a Deep Convolutional Neural Network (DCNN) for fire detection, demonstrating superior performance compared to traditional CNN models. To forecast the evolving trends of smoke and enhance smoke segmentation performance, Cheng et al. (2019) introduced a smoke detection method based on Deeplabv3+ and GAN (Generative Adversarial Network). Given the substantial variations in visual features of flame and smoke across different scenes, it is typically necessary to have extensive, high-quality training data to empower deep learning models with strong generalization performance (Torralba and Efros 2011). Current publicly available datasets in fire detection research are tailored to single sensors (space-based/air-based/ground-based), designated computer vision tasks (classification/object detection/segmentation), or simplified and similar scenes (indoor/outdoor). These datasets exhibit constraints, including

a limited number of training samples, consistent image dimensions or resolution, reliance on a sole data origin, and similar scenarios. Simultaneously, the potential application of space-air-ground integrated networks in the fire detection domain has been demonstrated (M. Wang et al. 2023). Deploying the same deep learning model collaboratively on such an integrated observation network could enhance early fire detection. Therefore, there is a critical requirement to construct a large-scale fire dataset containing a substantial number of diverse flame or smoke images from different origins. This dataset should be generated according to standardized principles and governed by FAIR (Findability, Accessibility, Interoperability, and Reusability) specifications (Geetha, Abhishek, and Akshayanat 2021).

This paper introduces a massive and diverse fire benchmark dataset designed for object detection tasks, encompassing flame and smoke images from various sources and scenes. The key contributions of this study can be summarized as follows: (1) We constructed a large-scale flame and smoke detection dataset, namely FASDD, which comprises over 120,000 samples. FASDD stands out as the most extensive and versatile fire object detection dataset, with unmatched intricacy in fire scenarios, considerable heterogeneity in feature distribution, and remarkable diversity in image size and shape. (2) The process of creating FASDD adheres to a unified and standardized data model, with annotations supplied in four widely accepted label formats, ensuring compatibility with various deep learning models. (3) We conducted extensive comparative experiments and performance evaluations using advanced object detection methods on FASDD. These experiments yield valuable insights for continually improving the performance of fire detection algorithms utilizing our benchmark dataset.

2. Related works

2.1. Existing fire datasets

Numerous fire detection datasets have been developed, some of which contain both flame and smoke data (Chino et al. 2015; Shamsoshoara et al. 2021), while others focus solely on smoke (Jakovcevic and Krstinic 2010; F. Yuan 2011; Q. Zhang et al. 2018) or flame (Dilshad et al. 2024; Dunning and Breckon 2018; Yar et al. 2023). These datasets consist of images or videos captured by surveillance systems or Unmanned Aerial Vehicles (UAVs). More recently, the increasing accessibility and affordability of satellite imagery have spurred research efforts in wildfire detection utilizing satellite data. For example, Ba et al. (2019) created a Moderate Resolution Imaging Spectroradiometer (MODIS) dataset for smoke scene classification. Yar et al. (2023) introduced the

Airborne Drone and Satellite Fire (ADSF) dataset, which covers a wider geographical area. Almeida Pereira et al. (2021) utilized patches from Landsat-8 imagery for wildfire area segmentation. These datasets primarily serve image scene classification and semantic segmentation tasks; however, there is still a scarcity of large-scale, multi-source, and heterogeneous fire benchmark datasets specifically designed for object detection tasks. Fortunately, existing fire datasets from spaceborne, airborne, and ground-based sensors offer valuable references for the development of a large-scale heterogeneous fire dataset with diverse scenes and rich characteristics.

2.2. Training data specification

Utilizing a standardized way to describe annotations is crucial for facilitating the sharing of datasets (Geetha, Abhishek, and Akshayanat 2021). Popular label formats in object detection tasks encompass JSON format designed in the Microsoft Common Objects in Context (COCO) dataset (Lin et al. 2014), XML format used in the Pascal Visual Object Classes (VOC) dataset (Everingham et al. 2015), and TXT format compatible with YOLO (Redmon et al. 2016) series models. The COCO dataset provides annotations for the entire dataset based on JSON files. It adopts the format $[x, y, w, h]$ to represent the coordinates of the top-left corner of the bounding box, as well as the width and height of the image. In contrast, the VOC dataset creates XML annotations for each image, utilizing the format $[x_1, y_1, x_2, y_2]$ to denote the coordinates of the top-left and bottom-right corners of the bounding box. Researchers of the YOLO algorithm annotate each image based on TXT files, employing the $[x, y, w, h]$ format to denote the normalized coordinates of the bounding box center, width, and height.

Additionally, a common language for describing various geospatial information is offered by the Spatio Temporal Asset Catalog (STAC). This process entails encoding a specific spatiotemporal asset as a GeoJSON entity, inclusive of date-time data and hyperlinks. The bounding boxes in STAC are depicted using either 2D or 3D geometries. A Training Data Markup Language (TDML) is proposed by Yue et al. (2022) to generate training samples for machine learning models. The TDML establishes a UML model and encoding methods that conform to the OGC standards baseline. It enables the seamless exchange and retrieval of geospatial machine learning training data on the web, aligning with prevalent JSON encoding practices. While retaining fundamental properties from other widely used data specifications, the TDML also includes comprehensive metadata to formalize the information model of training data. Datasets

generated using these standardized data specifications are poised to receive wider recognition and adoption within the deep learning community.

3. Dataset production

Given the constraints posed by the current fire datasets, both in terms of quantity and suitability for visual tasks, the FASDD is designed to be a large-scale dataset sourced from multiple sensors, featuring diverse resolutions, reflecting the complexity and generality of scenes, and adhering to standardized specifications. Figure 1 outlines the workflow for creating FASDD_CV, FASDD_UAV, and FASDD_RS, encompassing key stages such as data collection, preprocessing, annotation, and quality control.

3.1. Data source

To construct the fire detection dataset for Computer Vision (CV) tasks, various data sources, encompassing existing open-access fire datasets, social media data, Computer Graphics (CG) paintings, and images obtained through web crawlers, were utilized to create FASDD_CV. On the one hand, we are currently integrating and filtering flame or smoke images from 10 open-access datasets (Chino et al. 2015; Dunning and Breckon 2018; Foggia, Saggese, and Vento 2015; Geng 2020; Jakovcevic and Krstinic 2010; Ko, Kwak, and Nam 2012; Namburu et al. 2023; Sharma et al. 2017; F. Yuan 2011; Q. Zhang et al. 2018). For the extensive collection of raw images and videos, a uniform and standardized data cleaning and annotation process was applied. On the other hand, images containing challenging samples, considered as negative examples, were gathered through web crawling. These challenging samples include objects prone to confusion with smoke, such as dark clouds, shadows, hair, and impervious surfaces, as well as objects prone to confusion with flame, such as lights, sunset glow, and reflective clothing. Additionally, due to the continued scarcity of freely accessible drone data, emphasis has been placed on utilizing existing open-source drone datasets to establish FASDD_UAV, a dataset designed for UAV fire detection tasks. This primarily involves incorporating FLAME (Shamsoshoara et al. 2021), FLAME2 (Chen et al. 2022), FIgLib (Braun 2020), and the Wildfire-NET dataset (Muksimova, Mardieva, and Cho 2022).

To construct the dataset FASDD_RS for RS fire detection and localization tasks, we selected ten representative regions as study areas to collect fire remote sensing images from around the world. These areas are mainly derived from the summary of regions prone to frequent wildfires in recent years by Hu et al. (2021), as illustrated in Figure 2. These regions are distributed across all continents,

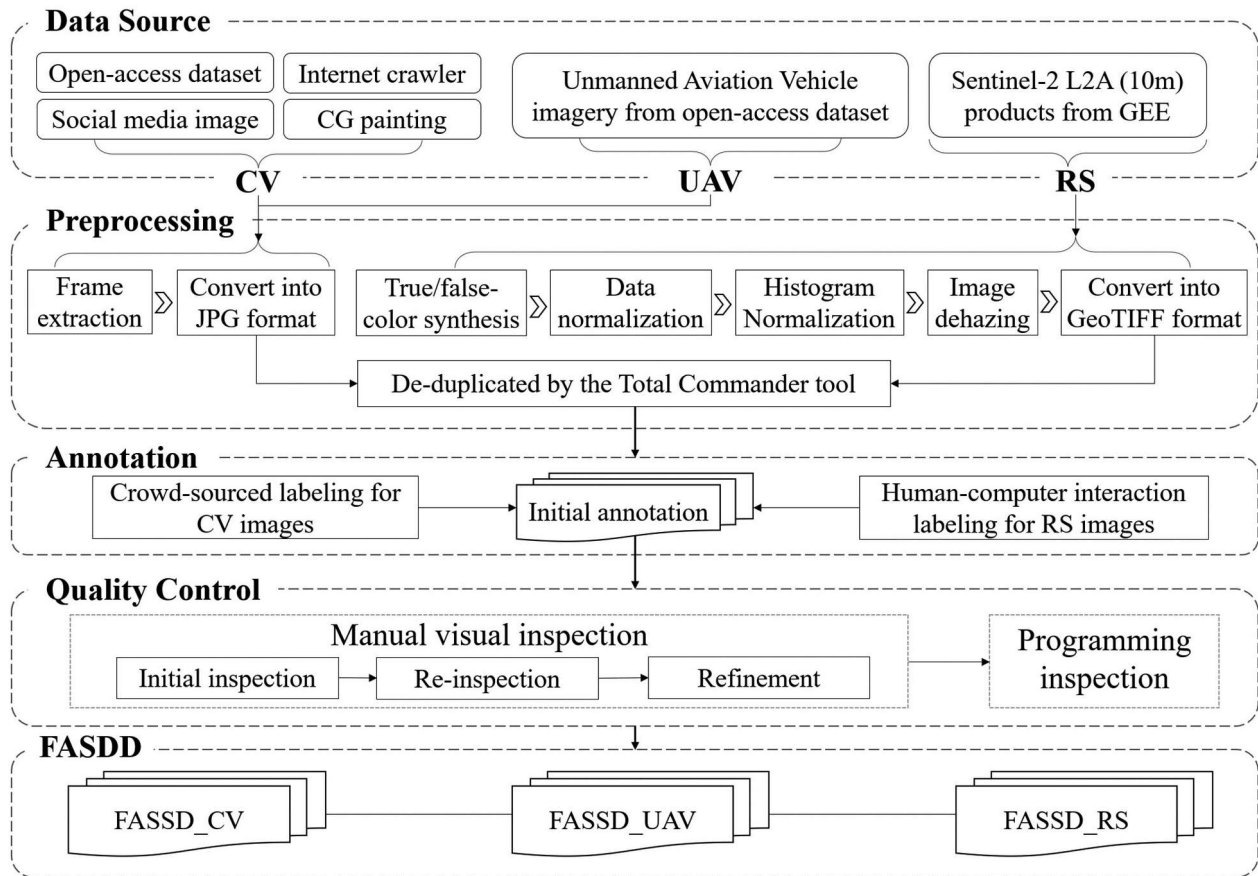


Figure 1. The workflow for creating FASDD.

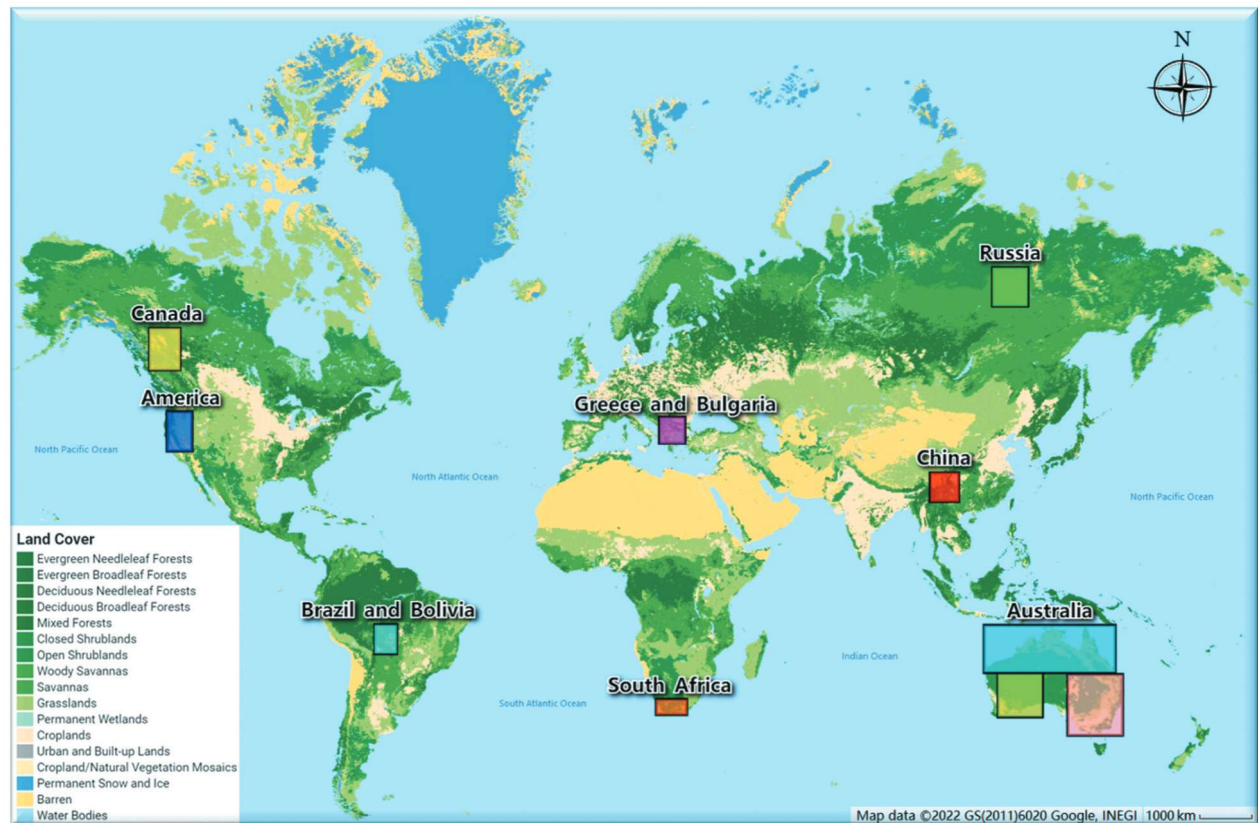


Figure 2. Representative regions of global fire incidents. The base map represents global land cover types from MODIS (Friedl and Sulla-Menashe 2019).

Table 1. Data sources for representative global fire incidents.

Region	Continent	Center coordinates	Time range	Data source	Resolution
Canada	North America	[−125.947, 53.846]	2019.01.01	Sentinel-2, L2A	10 m
America	North America	[−121.390, 39.813]	–2022.12.31		
Brazil and Bolivia	South America	[−60.616, −15.624]			
Greece and Bulgaria	Europe	[23.166, 38.010]			
South Africa	Africa	[22.431, −33.870]			
China	Asia	[101.465, 28.045]			
Russia	Europe	[119.99, 65.47]			
Australia	Oceania	[136.089, −15.676]			

excluding Antarctica, encompassing Canada and the United States in North America, Brazil and Bolivia in South America, Greece and Bulgaria in Europe, South Africa in Africa, China in Asia, Russia in Eurasia, and Australia in Oceania. To encompass the above fire-prone regions, Sentinel-2 imagery with a 10 m resolution has been gathered from 2019 to 2022. The utilization of Sentinel-2 data has been prevalent in numerous fire detection studies (Gargiulo et al. 2019; Hu, Ban, and Nascetti 2021). A summary of the remote sensing data sources utilized in this study is presented in Table 1. FASDD_RS is generated using atmospherically corrected Sentinel-2 L2A (Level 2A) products. We conducted data filtering based on cloud cover of less than 1%, and ultimately adjusted the RS image sizes to approximately 1000×1000 pixels. It is noteworthy that during the processing of satellite imagery data, spatial information such as longitude, latitude, and projection was preserved. This allows for the localization and tracking of forest fires within remote sensing images.

3.2. Data preprocessing

To maintain the uniformity and standardization of FASDD, preprocessing steps that ensure consistency need to be applied to the source data before annotating. For video data, the extraction of key frames is carried out, and images are sampled at intervals of 30 frames to introduce variation between samples. Moreover, all CV and UAV images, including those extracted from videos, undergo conversion into JPEG format files. Remote sensing imagery necessitates additional processing steps, encompassing band fusion, data normalization, and de-hazing of images. True-color image dataset FASDD_RS (RGB) for smoke annotation and false-color image dataset FASDD_RS (SWIR) (utilizing two SWIR bands and the Red band, i.e. Band 12, 11, and 4) for flame interpretation are synthesized from multispectral images. We conducted a bit-depth conversion process on all remote sensing images, ensuring that the pixel values in all images were normalized within the range of 0–255. This allows the produced remote sensing data can be seamlessly integrated with images from the CV or UAV domains for training deep learning models for

fire detection. Subsequent procedures involve histogram normalization and dehazing to fine-tune color components and enhance image clarity. The resultant remote sensing images are stored in GeoTIFF format files. Additionally, we provide a distinct dataset named FASDD_RS (All bands), containing RS images with all bands, for comparative experiments with the FASDD_RS (RGB) dataset, which exclusively uses RGB bands. Finally, to ensure the differentiation and uniqueness of image features within the dataset, all duplicate images are eliminated based on the Total Commander tool (Total Commander 2024).

3.3. Data annotation

In the image annotation tasks of the CV and UAV domains, we distribute all selected images to over 70 volunteers comprised of experts in fire detection for collaborative annotation in a crowd-sourced manner. We annotate the images offline using the Labelling tool, employing non-rotated minimum bounding rectangles to outline flame and smoke objects in the images. Additionally, we have established some fundamental annotation guidelines to standardize the annotation process, as shown in Figure 3. These annotation guidelines can be summarized as follows:

- (1) If a fire object is partially obstructed but visibly connected, it is treated as a distinct entity;
- (2) Clusters of multiple small objects are identified as a single entity;
- (3) Fire objects showing distinct colors are treated as separate entities;
- (4) Fire objects smaller than 10×10 pixels and lacking clear characteristics will be disregarded;
- (5) Fire objects with noticeable shape or texture features, even if smaller than 10×10 pixels, should not be discarded;
- (6) Reflections of fire objects on smooth surfaces, such as water, are excluded unless they preserve the shape and texture;
- (7) Images with dimensions less than 48×48 pixels that are challenging to interpret should be removed.

In the annotation tasks of remote sensing imagery, a semi-automatic approach is utilized for annotation,

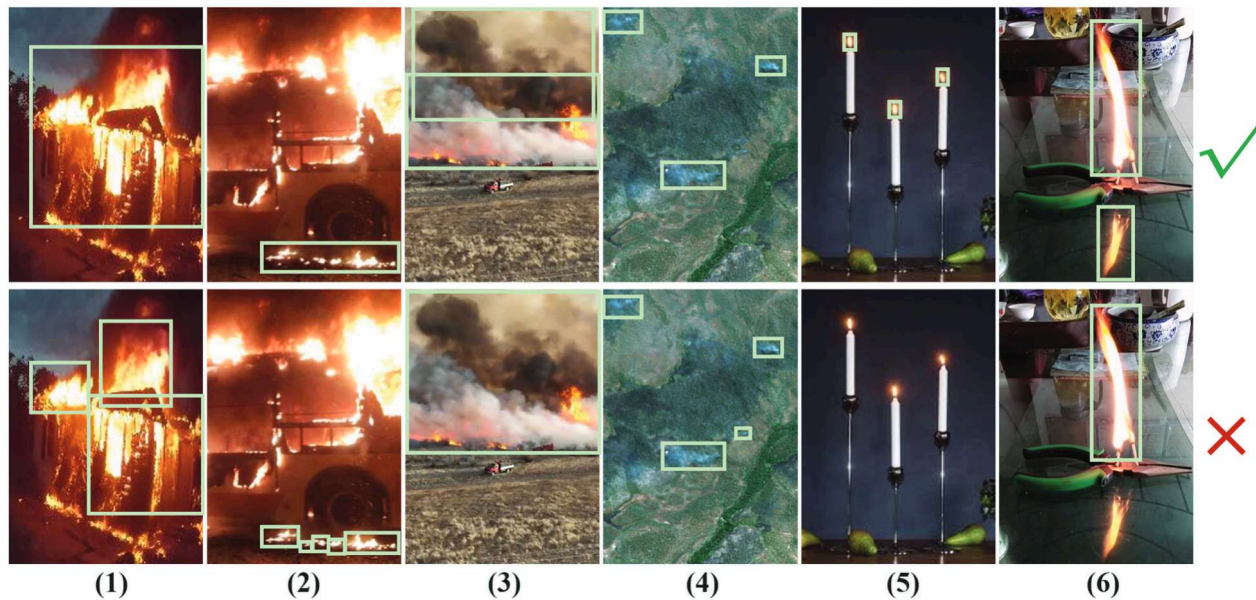


Figure 3. Annotation guidelines for the FASDD.

incorporating human-computer interaction. Initially, a flame and smoke detection model trained on existing FASDD_CV is employed to predict semantic annotations of RS images. Images with inference confidence scores exceeding 80% are selected as target images, which are potential training samples likely to contain flame and smoke objects. Subsequently, these target images are further categorized into positive samples containing flame and smoke, and negative samples resembling flame or smoke, through manual visual interpretation. All target images are then distributed to a small group of trained annotators to generate annotation files. Finally, smoke objects in FASDD_RS (RGB) and flame objects in FASDD_RS (SWIR) are annotated through human-machine interaction.

The FASDD dataset is primarily designed for object detection tasks, with flame and smoke entities in images labeled as “fire” and “smoke” respectively. FASDD offers annotation in four different formats: VOC, YOLO, COCO, and TDML. As all images belong to one of

four categories – “fire”, “smoke”, “FireAndSmoke”, and “NeitherFireNorSmoke” - the category name will be incorporated into each image filename as a prefix. This naming convention allows FASDD to be utilized for training fire scene classification models as well.

3.4. Quality control

To guarantee the reliability and reusability of FASDD, a series of quality assurance protocols are devised, as depicted in Figure 4. Firstly, three-stage manual inspection processes are established after the acquisition of the initial annotations, namely the initial inspection, re-inspection, and refinement stages, aimed at rectifying uncertain data. During the initial inspection stage, pairs of annotators are grouped to cross-validate and amend annotation files, facilitating the identification of inconsistent labels among different annotators and mitigating cognitive biases inherent in crowdsourced annotations. During the re-inspection stage, addressing issues identified during the initial inspection, a specialized panel of

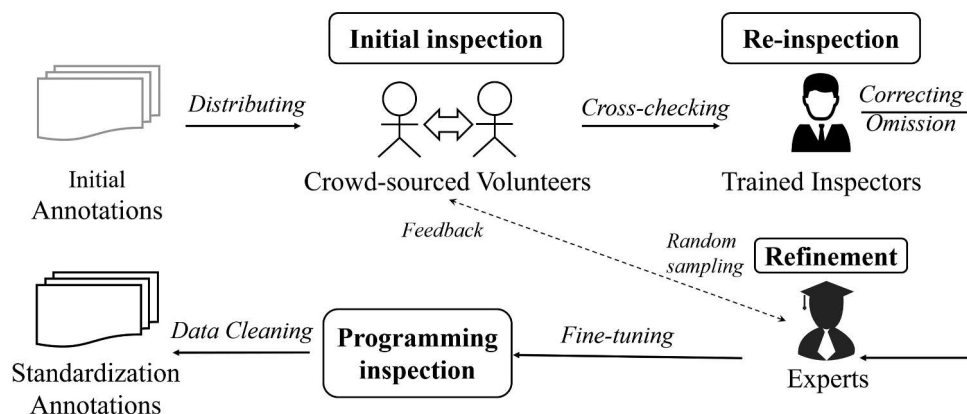


Figure 4. The quality control flowchart.

quality inspectors undergo training to review outcomes from the initial inspection phase, eliminating omission errors and refining the position, size, and category of bounding boxes. During the refinement stage, proficient experts are invited to randomly inspect the quality of the initial inspection results, and any data issues found will be provided as feedback to the initial inspectors for modification. More importantly, the experts will conduct final refinements and improvements to the re-inspection results. These experts comprise researchers from multiple fields including remote sensing, fire detection, optoelectronic information processing, and computer vision, with extensive experience in fire detection research and a profound academic background in artificial intelligence.

In addition, following manual inspection procedures, we implement a programming inspection process. This process employs an annotation verification code to carry out the final data-cleaning task across all labels. The code can automatically identify empty, duplicated, or out-of-range bounding boxes, as well as misclassified or misspelled labels. This aids in averting errors or omitted values in labels that may not be easily detected through manual visual inspection alone. With the dual assurance of manual and programmatic inspection procedures, we can better ensure the consistency and standardization of the produced dataset.

3.5. Data characteristics

FASDD encompasses images of fire, and smoke, as well as potentially confusing non-fire or non-smoke scenes captured at various scenes (indoors/outdoors), different perspective distances (near/far), different light conditions (daytime/nighttime), and derived from multi-source heterogeneous visual sensors

(video surveillance, drones, and spaceborne multi-spectral sensors). The example images are depicted in Figure 5. FASDD is comprised of three sub-datasets: a CV dataset (FASDD_CV), a UAV dataset (FASDD_UAV), and an RS dataset (FASDD_RS). It is noteworthy that these subdatasets can operate independently. The FASDD_CV dataset can provide crucial data support for urban firefighting. Due to the wide coverage of remote sensing images in FASDD_UAV and FASDD_RS, they are particularly suitable for specialized applications in forest fire detection. Additionally, merging these datasets into a unified repository for collaborative use is also feasible. First, the diverse training data sourced from multiple channels aims to streamline the development of a robust fire detection model with enhanced performance and generalizability, particularly across specific scenarios. Second, FASDD enables seamless deployment of a single model across spaceborne, airborne, and ground-based sensors, facilitating the establishment of an integrated fire detection network spanning multiple domains.

Table 2 presents a comprehensive summary of FASDD, comprising 122,634 samples, with 70,581 annotated as positive and 52,073 as negative samples. The dataset includes 113,154 flame instances and 73,072 smoke instances. FASDD_CV (95,314 samples) targets general computer vision, FASDD_UAV (25,097 samples) is captured by UAV, and FASDD_RS (2223 samples) is derived from satellite imagery. FASDD_CV features 73,297 fire objects and 53,080 smoke objects with varying image sizes (78 to 10,600 pixels in width, 68 to 8858 pixels in height) and aspect ratios (1:6.6 to 1:0.18). FASDD_UAV includes 36,308 fire instances and 17,222 smoke instances, with aspect

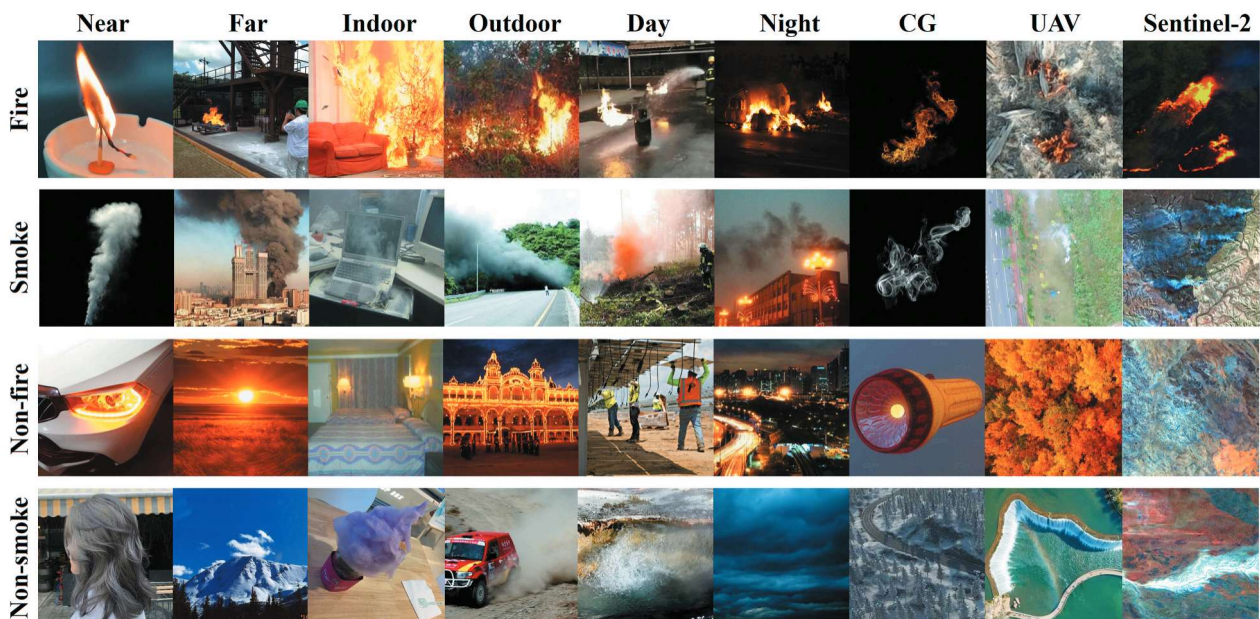


Figure 5. Sample images from FASDD.

Table 2. The composition and characteristics of the FASDD.

Datasets	Images	Train	Val	Test	Positive samples	Negative samples	Flame objects	Smoke objects	Width range	Height range	Aspect ratios
FASDD_CV	95,314	47,660	31,770	15,884	56,115	39,199	73,297	53,080	78–10,600	68–8858	1:6.6–1:0.18
FASDD_UAV	25,097	12,551	8365	4181	13,111	11,986	36,308	17,222	1280–3072	720–2048	4:3–16:9
FASDD_RS	2223	1112	741	370	1335	888	3549	2770	1000×1000	1000×1000	1:1
FASDD	122,634	61,323	40,876	20,435	70,561	52,073	113,154	73,072	78–10,600	68–8858	1:6.6–1:0.18

ratios mostly between 4:3 and 16:9. FASDD_RS has 2770 smoke instances and 3549 flame instances, with remote sensing image sizes around 1000×1000 pixels.

Compared with existing fire detection datasets, FASDD demonstrates the following notable characteristics:

- (1) Larger-scale Dataset: FASDD encompasses over 120,000 images and more than 180,000 instances meticulously annotated using bounding boxes. As far as our knowledge extends, it stands as the largest, most comprehensive, and open-access object detection dataset for fire detection.
- (2) Various Sample Characteristics: The proposed FASDD dataset showcases a wide array of variations in terms of image size, resolution, light intensity, distance perspectives, sensor, scene, and data source. These diverse image characteristics are poised to enhance model robustness.
- (3) Significant Intra-Class Diversity and Some Inter-Class Resemblance: Given the growth, irregularity, color variation, and complex illumination environment of fire data, flame or smoke objects exhibit variations in size, shape, and color. Additionally, there exist resemblances between flames and smoke, such as the reddish hue of smoke illuminated by flames resembling flames themselves.
- (4) Lots of Small Flame and Smoke Objects: The detection of diminutive objects, particularly flame entities from remote sensing imagery and distant wildfire scenes, poses a formidable challenge in deep learning object detection tasks. FASDD encompasses a substantial number of such small-scale flame and smoke entities.
- (5) Georeferenced RS Samples: In contrast to conventional computer vision datasets, FASDD incorporates a significant number of georeferenced images. The spatial information embedded within remote sensing images can facilitate the timely detection and location of the fire.

4. Evaluation and discussion

4.1. Experiment configuration

In our experiment, we follow the established random partitioning strategies (Agrawal, Girshick, and Malik 2014; Ma et al. 2022; Xia et al. 2018). The Swin Transformer (Liu et al. 2021), showcasing State-Of-The-Art (SOTA) performance on the COCO dataset, serves as the baseline model for performance evaluation. Two training modes are employed: training the Swin Transformer (ST) model from scratch and training ST using the pre-trained model. To ensure a fair performance comparison, all models share the same configuration, including 36 epochs, a batch size of 2, and the AdamW optimizer with a learning rate of 0.0001 and weight decay of 0.05. Multi-scale training is applied to all models, with the image size ranging from 1333×480 to 1333×800 . All models undergo training, validation, and testing on a single NVIDIA GeForce RTX 3090 with 24 GB video memory.

4.2. Performance metrics

We employ four metrics to quantitatively assess the model performance: Precision (P), Recall (R), Average Precision (AP), as well as mean Average Precision (mAP). In object detection tasks, precision refers to the ratio of correctly identified positive detections to the total number of positive detections made by the model. Recall, on the other hand, represents the ratio of correctly identified positive detections to the total number of actual positive instances in the dataset. AP represents the Area Under Curve (AUC) of the Precision-Recall curve for each class in the dataset, and mAP denotes the average AP value across all categories. We utilize mAP50, a common metric derived from mAP, as the primary measure for assessing model performance. mAP50 corresponds to mAP when the Intersection over Union (IoU) between predicted and actual bounding boxes exceeds or equals 50%. The computation of Precision and Recall is delineated in Equations (1) and (2):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

where TP denotes the box count where the IoU between predicted results and actual bounding boxes is greater than or equal to 50%. FP denotes the box count where the IoU between predicted results and actual bounding boxes is below 50%. FN denotes the count of actual bounding boxes that were not detected.

4.3. Performance evaluation

The Swin Transformer (ST) models were trained on FASDD_CV, FASDD_UAV, and FASDD_RS training sets, and their accuracy was assessed on corresponding validation and test sets (refer to Table 3 for results). Specifically, the ST model was trained directly from scratch on the dataset, while the CVPT-ST model was Pre-Trained on the FASDD_CV dataset and then further fine-tuned on the pre-trained model. The AllB-ST model was also trained directly from scratch on the dataset, but unlike other models, it utilized remote sensing images containing All spectral Bands (AllB). The ST model achieved excellent performance on FASDD_CV with a mAP50 of 69.8% (validation) and 78.8% (test). The Recall for flame and smoke objects exceeded 96% and 91%, respectively. On FASDD_UAV, both ST and CVPT-ST models achieved over 91.5% mAP50, 93% Recall for flame, and 96.6% Recall for smoke. The evaluation on FASDD_RS explored different band combinations. The ST model on FASDD_RS (RGB) achieved mAP50 of 68.4% (validation) and 68.8% (test). The CVPT-ST model improved mAP50 by 2.0% (validation) and 3.2% (test), reaching the highest accuracy of 72%, with a 91.8% Recall for smoke objects. For

FASDD_RS (SWIR), the ST model achieved mAP50 of 80.7% (validation) and 81.7% (test), with 94.9% and 95.4% Recall. The CVPT-ST model slightly improved mAP50 on validation and test sets while maintaining a similar Recall compared to the ST model.

Additionally, we trained a model from scratch on satellite imagery incorporating both fire and smoke labels across all bands. The accuracy assessment on FASDD_RS (All Bands) showed that the AllB-ST model performed well in detecting flame and smoke objects, despite a decrease in accuracy compared to models specialized for smoke (FASDD_RS RGB) and flame (FASDD_RS SWIR) detection. Multispectral data proved beneficial for simultaneous flame and smoke detection by extracting richer feature information from multiple bands. However, this advantage comes with increased computational demands and potential accuracy loss due to redundant band interference. Notably, the “pre-training + fine-tuning” approach contributed to improved model performance, highlighting the significance of large-scale heterogeneous data in fire detection models.

To validate model generalization, we assessed the inference accuracy on FASDD_CV, FASDD_UAV, and FASDD_RS (RGB) using models trained on FASDD, as shown in Table 4. Employing the Swin Transformer model pre-trained on ImageNet-1K as a baseline, we fine-tuned it on FASDD, referred to as ImgPT-ST. On the FASDD_CV test set, ImgPT-ST outperformed models solely trained on FASDD_CV, achieving an 84.9% mAP50 (6.1% improvement), a 1.9% improvement in AP (fire), an 8.8% improvement in AP (smoke), and an average Recall of 95.25% for fire and smoke objects. For the FASDD_UAV test set, ImgPT-ST showed

Table 3. Accuracy evaluation based on the Swin Transformer.

Dataset	Model	Epoch	Validation (%)					Test (%)				
			AP (fire)	AP (smoke)	mAP50	Recall (fire)	Recall (smoke)	AP (fire)	AP (smoke)	mAP50	Recall (fire)	Recall (smoke)
FASDD_CV	ST	36	65.0	74.6	69.8	96.0	91.5	81.5	76.1	78.8	96.3	91.2
FASDD_UAV	ST	36	87.7	96.0	91.8	93.6	97.1	87.1	96.0	91.6	93.1	97.3
	CVPT-ST	36	87.8	96.0	91.9	93.4	97.0	87.6	95.5	91.5	93.0	96.6
FASDD_RS (RGB)	ST	36	–	68.4	68.4	–	89.0	–	68.8	68.8	–	88.8
	CVPT-ST	36	–	70.4	70.4	–	89.9	–	72.0	72.0	–	91.8
FASDD_RS (SWIR)	ST	36	80.7	–	80.7	94.9	–	81.7	–	81.7	95.4	–
	CVPT-ST	36	82.2	–	82.2	94.8	–	82.2	–	82.2	94.8	–
FASDD_RS (All bands)	AllB-ST	36	77.8	66.7	72.3	93.3	84.7	78.5	66.7	72.6	93.6	85.4

Table 4. Performance evaluation on each subset employing models trained on FASDD.

Test Dataset	Training Dataset	Model	Test (%)				
			AP (fire)	AP (smoke)	mAP50	Recall (fire)	Recall (smoke)
FASDD_CV	FASDD_CV	ST	81.5	76.1	78.8	96.3	91.2
	FASDD	ImgPT-ST	83.4	86.4	84.9	96.1	94.4
FASDD_UAV	FASDD_UAV	ST	87.1	96.0	91.6	93.1	97.3
	FASDD	ImgPT-ST	83.5	95.9	89.7	93.6	98.6
FASDD_RS (RGB)	FASDD_RS (RGB)	ST	–	68.8	68.8	–	88.8
	FASDD	ImgPT-ST	–	74.0	74.0	–	93.0

comparable AP and mAP accuracies to ST and CVPT-ST models, attaining an 89.7% mAP50 and a 96.1% average Recall. The ImgPT-ST model on FASDD outperformed models trained solely on FASDD_RS (RGB) for the RS dataset, achieving a 74.0% mAP50 and a 93% Recall for smoke objects. Results from FASDD_CV and FASDD_RS (RGB) tests indicated ImgPT-ST's superior ability to leverage common features learned from complex scenes, showcasing substantial advantages in challenging sub-datasets. Therefore, for fire detection tasks spanning across CV, UAV, and RS domains, FASDD can be utilized to develop deep learning models compatible with spaceborne, airborne, and terrestrial sensors, enabling collaborative deployment and application of fire detection models within space-air-ground integrated systems. This approach ensures consistency and compatibility across diverse sensor types, enabling effective fire detection across multiple domains. Furthermore, we have conducted extensive validation to establish the broad applicability of the FASDD, using various neural network architectures. The corresponding experimental result is detailed in the [Appendix](#).

4.4. Visual results

Figures 6–8 display the prediction results of Swin Transformer models on the CV, UAV, and RS datasets, respectively. Here, “XX_Both” refers to images in the “FireAndSmoke” category, while “XX_None” refers to images in the “NetherFireNorSmoke” category. “Raw” represents the input image used for model

prediction, and “Label” represents the true label in annotations.

In [Figure 6](#), the ST model accurately predicts CV_Fire (flames only) and CV_Both (containing flames and smoke). However, in CV_Smoke (smoke only), where only smoke is present, the ST model effectively detects the large smoke area but mistakenly identifies flames as headlights with a confidence level of 51%. Conversely, in CV_None (lacking flames and smoke), the ST model incorrectly identifies an orange light as a flame object with 51% confidence, possibly due to the Swin Transformer's global attention bias. These instances highlight the complexity and challenges of our dataset.

In [Figure 7](#), the ST model produces duplicate predictions for UAV_Fire by assigning two different-sized label boxes to the same object. In contrast, the CVPT-ST model accurately predicts flames with 96% confidence. For UAV_Both, both ST and CVPT-ST models successfully predict without falsely detecting the red sun. In UAV_Smoke, the ST model aligns well with the true label, while the CVPT-ST model detects an unlabeled small flame. In UAV_None, the ST model mistakenly identifies a cloud as smoke, while the CVPT-ST model performs commendably. The CVPT-ST model, pre-trained on FASDD_CV, captures detailed features and demonstrates superior generalization.

In [Figure 8](#), the ST model generates overlapping label frames for RS_Smoke, but both CVPT-ST and AllB-ST models predict the same smoke object

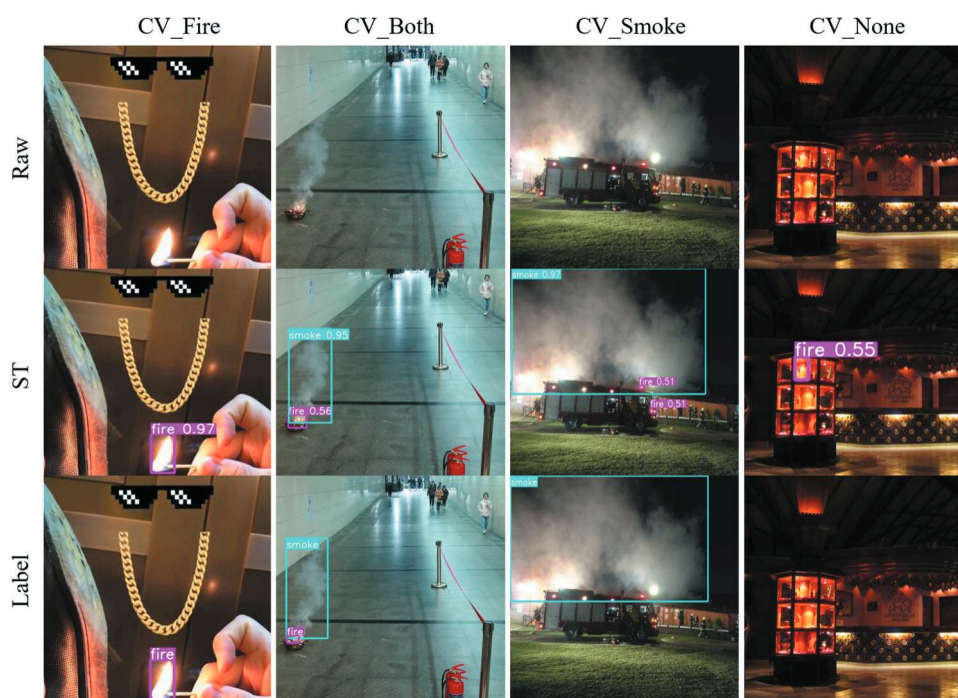


Figure 6. Visual results of the swin transformer model on CV example images.

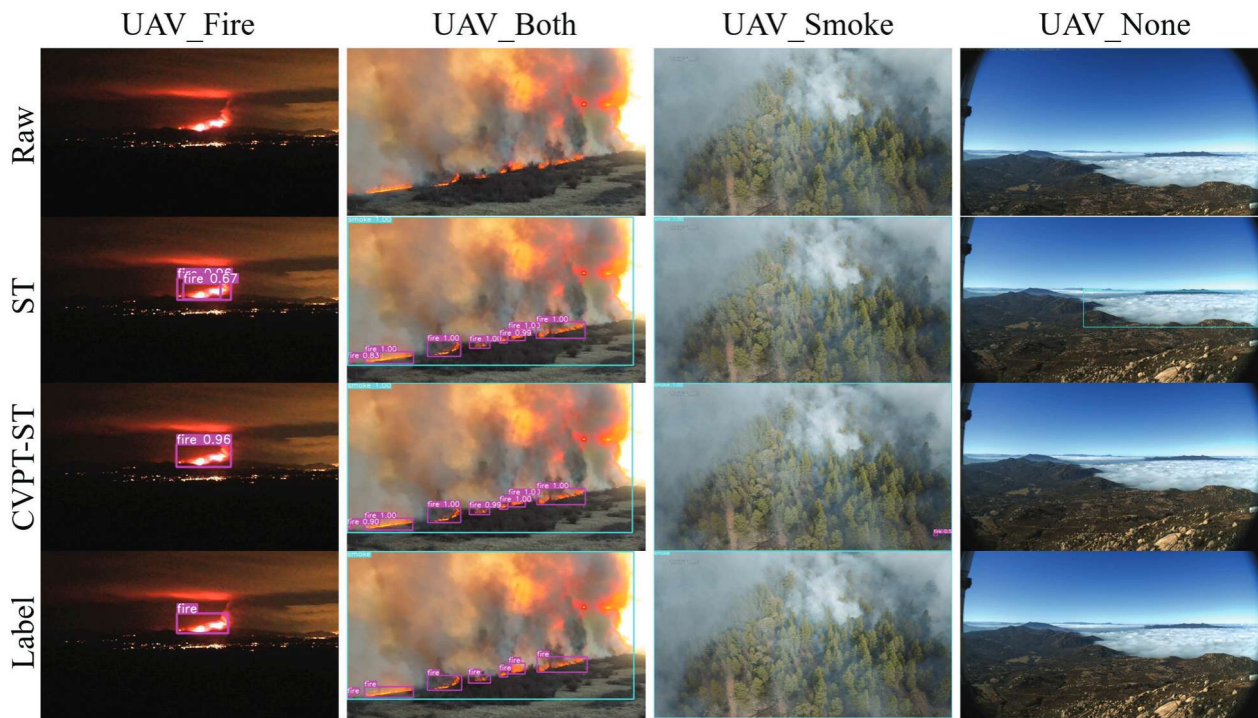


Figure 7. Visual results of swin transformer models on the UAV example images.

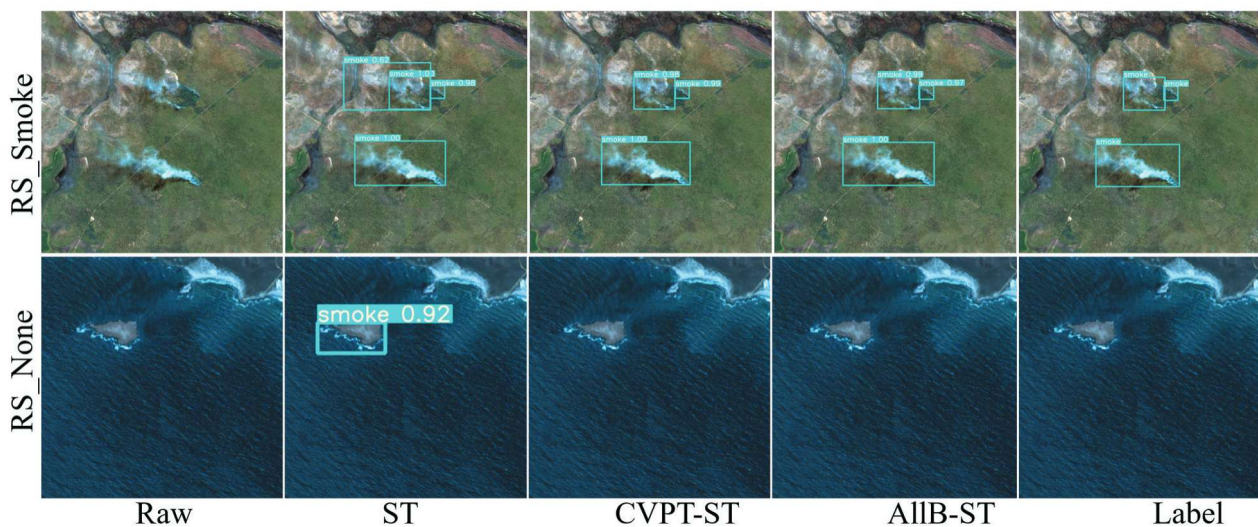


Figure 8. Prediction results of the RS example images.

accurately. In RS_None, the ST model erroneously detects curved waves as smoke, while CVPT-ST and AllB-ST models demonstrate equally reliable predictions. These findings highlight that the CVPT-ST model, following pre-training on FASDD_CV, excels in discriminating between positive and negative samples, showcasing performance comparable to the AllB-ST model.

4.5. Application for fire localization

The aforementioned methods are implemented to conduct wildfire localization tasks on remote sensing images with spatial references. The localization

accuracy of models is validated using the position coordinates of the predicted results. Initially, the RS_Smoke sample image (Figure 8) is transformed into a WGS84 coordinate system. Subsequently, inferences are executed on the georeferenced RS_Smoke image using the ST, CVPT-ST, and AllB-ST object detection models, respectively. Finally, the location of the predicted boxes is transformed into latitude and longitude coordinates of the top-left and bottom-right corners to compare the wildfire localization capabilities of different models.

Table 5 presents a comparison of the location coordinates, centroid distance deviation, and IoU values between the predicted results and actual bounding boxes of the RS_Smoke image. A hyphen “-” indicates the absence of a value in the respective label box location.

Table 5. Comparison of location between predicted and actual bounding boxes.

Box	Model	Top Left Coordinate	Bottom Right Coordinate	Centroid Distance Bias (m)	IoU
Box1	ST	[129.9506, -13.9988]	[129.9842, -14.0153]	205.91	0.81
	CVPT-ST	[129.9515, -13.9993]	[129.9843, -14.0151]	158.11	0.84
	AllB-ST	[129.9503, -13.9993]	[129.9842, -14.0153]	216.62	0.83
	Ground Truth	[129.9535, -14.0002]	[129.9846, -14.0158]	0.00	1.00
Box2	ST	[129.9786, -13.9784]	[129.9840, -13.9830]	55.90	0.79
	CVPT-ST	[129.9789, -13.9786]	[129.9840, -13.9830]	45.28	0.83
	AllB-ST	[129.9785, -13.9783]	[129.9840, -13.9834]	42.72	0.83
	Ground Truth	[129.9788, -13.9786]	[129.9840, -13.9838]	0.00	1.00
Box3	ST	[129.9634, -13.9749]	[129.9786, -13.9870]	65.19	0.89
	CVPT-ST	[129.9635, -13.9752]	[129.9787, -13.9870]	49.24	0.92
	AllB-ST	[129.9631, -13.9752]	[129.9788, -13.9869]	67.08	0.89
	Ground Truth	[129.9638, -13.9752]	[129.9792, -13.9874]	0.00	1.00
Box4	ST	[129.9465, -13.9698]	[129.9788, -13.9873]	–	–
	CVPT-ST	–	–	–	–
	AllB-ST	–	–	–	–
	Ground Truth	–	–	–	–

The CVPT-ST and AllB-ST models successfully identify all bounding boxes closely matching the label coordinates. However, the ST model erroneously detects an additional bounding box not present in the label. When evaluating centroid distance, the CVPT-ST model demonstrates the smallest bias compared to the other models, particularly for the first and third bounding boxes. Regarding IoU, both the CVPT-ST and AllB-ST models yield favorable outcomes, achieving an IoU exceeding 83% for all prediction boxes. Remarkably, the CVPT-ST model surpasses the others, achieving the highest IoU with the actual bounding boxes. Specifically, it achieves a 92% IoU with the ground truth for the third bounding box. These comparison results underscore the accuracy and efficacy of the CVPT-ST model trained on FASDD_RS in precisely detecting and tracking wildfire positions in remote sensing images. This holds significant reference value for the detection and monitoring of large-scale urban or forest fires based on remote sensing satellite imagery.

5. Data availability

FASDD can be accessed without charge on the Science Data Bank website at <https://doi.org/10.57760/sciencedb.j00104.00103> (M. Wang et al. 2024). The dataset is furnished in five compressed archives: FASDD.zip, FASDD_CV.zip, FASDD_UAV.zip, FASDD_RS.zip, and FASDD_RS_SWIR.zip. Each zipped file includes two directories: “images” for the storage of raw images and “annotations” for the storage of object detection labels. The label files in the “annotations” directory are available in four formats: VOC, COCO, YOLO, and TDML. The dataset is randomly partitioned into training, validation, and test subsets, with proportions of 1/2, 1/3, and 1/6, respectively, within each label format. The image and label designations are prefixed with “Fire,” “Smoke,” “FireAndSmoke,” and “NeitherFireNorSmoke,” denoting distinct categories of images for scene classification tasks.

6. Conclusions

This paper introduces an accessible Flame and Smoke Detection Dataset, FASDD, which comprises more than 120,000 instances. As far as our knowledge extends, FASDD stands as the largest fire detection dataset, characterized by a diverse array of real-world scenarios, considerable heterogeneity, and notable variations in feature distribution. The annotation files are available in four formats, facilitating the use of the dataset as off-the-shelf training samples for deep learning models. Leveraging the proposed dataset, we carry out comprehensive evaluations and comparisons using Swin Transformer object detection models. The outcomes reveal that Swin Transformer models achieve remarkable performance, exhibiting mAP50 scores of 84.9%, 89.7%, and 74.0%, along with average Recall rates of 95.25%, 96.1%, and 93.0% on the test sets of FASDD_CV, FASDD_UAV, and FASDD_RS, respectively. Furthermore, in applications related to wildfire localization, the CVPT-ST model, trained on FASDD_RS, produces localization results in close agreement with the annotations, underscoring its crucial role in early warning and detection of urban and forest fires. The FASDD dataset stands as a benchmark for the development of advanced fire detection models applicable to ground-based, airborne, and spaceborne systems in the space-air-ground integrated observation network. These models can be tailored to regional and global-scale fire scenarios, providing valuable insights for governmental decision-making and fire-fighting operations. In future dataset optimization efforts, we will employ active learning techniques to annotate the most valuable and challenging hard samples, aiming to reduce errors and biases on negative samples. Additionally, we will supplement the dataset with data of different scales, such as the remote sensing images with varying resolutions and drone images captured at the different altitudes. Furthermore, considering the introduction of data augmentation techniques like random cropping, mosaicing, and scaling to enhance dataset diversity, we will facilitate the

model in learning richer features, thereby strengthening its adaptability and generalization to targets of different scales.

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Appendix

To comprehensively validate the universality and reliability of our dataset, we conducted extensive benchmark testing on FASDD using four state-of-the-art deep learning models. These models followed the previous experimental setup, except for utilizing a larger batch size of 8. These models encompass those based on CNN architecture (YOLOv5× (Jocher et al. 2021) and InternImage (W. Wang et al. 2023)) as well as those rooted in transformer architecture (DETR (Carion et al. 2020) and Swin Transformer). Table A1 showcases the accuracy evaluation results of models with different network architecture paradigms on both the FASDD validation and test sets.

Table A1 illustrates the consistent stability in performance exhibited by both CNN-based and transformer-based models across the validation and test sets of FASDD. Notably, the top-performing models identified through validation set filtering consistently achieve a mAP50 exceeding 80.4% on the test set. Except for the YOLOv5× model, recall rates for other models range from 92.7% to 97.2%. Furthermore, most models, excluding the DETR model, achieve a mAP50 of approximately 84%. Among these models, the advanced InternImage and Swin Transformer demonstrate comparable accuracy, achieving a noteworthy 78.1% mAP50 on the validation set and surpassing 84% mAP50 on the test set. These comparative results underscore the dataset's broad applicability and reliability, accommodating models across various architectural paradigms and algorithmic ideas.

Table A1. Accuracy evaluation results on FASDD for models with different basic architectures.

Dataset	Batch size	Epoch	Network architecture paradigm	Model	Validation (%)					Test (%)				
					AP (fire)	AP (smoke)	mAP50	Recall (fire)	Recall (smoke)	AP (fire)	AP (smoke)	mAP50	Recall (fire)	Recall (smoke)
FASDD	8	36	CNN	YOLOv5x	73.8	77.2	75.5	75.0	61.0	84.4	83.1	83.7	83.8	66.2
				InternImage	74.0	82.2	78.1	90.7	90.8		85.4	84.0	93.6	92.7
			Transformer	DETR	68.8	80.3	74.5	89.9	96.2	82.6	83.8	80.4	92.7	97.2
				Swin Transformer	72.8	83.4	78.1	91.5	92.8	77.0	86.3	84.2	94.1	94.1
										82.0				